JOSHUA STAFFORD

London, SE20 7ST | 07944536568 | jegstafford@gmail.com

Summary

Results-driven, outgoing software engineer with a proven track record of delivery. 2.5 years of professional industry experience. Flexible and willing to relocate for work. Long history of hospitality jobs prior to professional career, happy to provide details if relevant.

Experience

Software Engineer **09/2022 to Current**
Viridien
Lead Developer on Kickoff API integral to Department

- API allows users to kickoff & monitor Machine Learning pipelines from both Spark and Argo
- Written in Python using FastAPI framework
- Integrated with Postgres DB for vital data retention.
- Containerised using Docker and deployed using Kubernetes
- Deployed both locally and to Azure
- Integrated with a UI written by the frontend developer.
- Version control done with GitLab
- Pipeline outputs are stored in Neo4j using GraphQL.
- RabbitMQ is used for information transfer from the pipeline to the API.

Developed JSON to Cypher API

- API compiles JSON to cypher query and runs against graph to return results
- Written in Python using FastAPI framework
- Takes dictionary of node names with requested filters and returnable attributes
- Uses breadth-first search to compute the shortest path through graph via relationships between nodes.
- Currently only integrated with Neo4j but working towards integration with ElasticSearch
- Deployed in multiple client environments using Kubernetes

Skills

- SQL
- Python
- FastAPI
- Docker
- Kubernetes
- Azure Webservices
- GraphQL
- Git
- Linux
- Spark
- RabbitMQ
- Kafka
- Postgres
- Agile programming
- Project Management
- API design

Education

Bachelor of Engineering: **Computer Science** **2022**
University of York York

- Graduated with a first class honours degree
- Dissertation in video classification using Convolutional Neural Networks
- President of Derwent Squash Team

A-levels **2019**
Uckfield Community College Uckfield

- Mathematics: A
- Computer Science: B
- Physics: B
- EPQ - Self-driving toy car: A*
- I modified the toy car with a raspberry pi, motor controller and webcam, implemented a feed-forward neural network from scratch in Python, and collected training data from the webcam using a remote-control UI implemented in Nodejs / javascript to control the car with my laptop arrow keys.